Deepak Mallampati

MACHINE LEARNING ENGINEER

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PROFESSIONAL SUMMARY

ML engineer with 4+ years shipping production AI systems across text, multimodal, and LLM domains. Builds classification, safety, and evaluation systems end-to-end — engineered content-safety controls that cut unsafe model outputs 40%+ and a structured LLM evaluation framework (RAGAS, BERTScore, custom metrics) that catches regressions before release. Strong in PyTorch/TensorFlow, agentic systems, model optimization, and MLOps (Docker, Kubernetes, CI/CD) across GCP, AWS, and Azure. US-based, authorized to work now.

CORE COMPETENCIES

Machine Learning • Classification & Abuse Detection • LLM Evaluation & Benchmarking • Content Safety • Agentic Systems • Model Optimization • MLOps • Multimodal

ML & Deep Learning: PyTorch, TensorFlow, Keras | NER, Text Classification, Summarization | Model Optimization | A/B Testing | Drift Detection | RAGAS Evaluation | BERTScore | Hugging Face Transformers | Fine-tuning (LoRA, QLoRA, PEFT)

Generative AI & LLMs: GPT-4o, GPT-5, Claude Sonnet, Gemini, LLaMA | LangChain/LangGraph | RAG Architecture | Multi-Agent Systems | Content Safety | Prompt Engineering

MLOps & Cloud: GCP Vertex AI, AWS (Bedrock, SageMaker, Lambda), Azure (OpenAI, Content Safety) | Docker, Kubernetes | CI/CD (Jenkins, Azure DevOps) | MLflow, Weights & Biases | Model Monitoring & Observability

Vector Search & Retrieval: FAISS, Pinecone, Weaviate | Semantic & Hybrid Search | Cross-Encoder Re-ranking | Semantic Caching

Languages & Tools: Python, JavaScript/TypeScript, Java, SQL | FastAPI | Git, Jira | Agile/Scrum

PROFESSIONAL EXPERIENCE

Progress Solutions — *Jul 2025 – Present*

AI Engineer | USA

- Architected production-grade **agentic systems** on a conductor–subagent orchestration framework, decomposing complex objectives into context-scoped subagents — reducing token consumption by **42%** while sustaining task accuracy across multi-step workflows, enabling large-scale automated experimentation.
- Established a structured **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency and accuracy dashboards to benchmark model iterations pre-deployment and surface regressions before release.
- Engineered high-performance **retrieval-augmented generation (RAG)** pipelines over large document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift relevance and materially reduce hallucination.
- Optimized inference cost and latency through **prompt compression, semantic caching, and model routing**, sustaining SLA targets while reducing per-query overhead.
- Built fault-tolerant automation infrastructure with scheduled batch orchestration and exponential-backoff retry logic, reducing failed production runs by **60%**.

Vanguard — *Jan 2024 – Jan 2025*

AI Engineer Intern | USA

- Engineered **safeguards and content-safety controls** for secure, policy-compliant AI responses, reducing unsafe outputs by **42%** while maintaining response quality.
- Optimized LLM prompts and **benchmarked GPT-4o, Claude Sonnet, and Gemini**, improving task success rates by **27%** on internal evaluation datasets and informing platform model-selection decisions.
- Developed and enhanced **AI agents and backend services** for Vanguard's internal AI platform across **25+ agents** and workflows, integrating with production data pipelines and observability tooling.
- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and supporting **100,000+ requests daily** across 20+ internal teams.

Kent State University
Oct 2023 – Jan 2024

ML & Gen AI Research Assistant | USA

- Built a **privacy-preserving ML pipeline** on clinical data (k-anonymity, l-diversity, Laplace differential privacy), reducing re-identification risk from 3.38% to **0.32%** while retaining **99%** of baseline predictive performance.
- Designed a composite **privacy–utility scoring framework** benchmarking six anonymization configurations to identify the optimal operating point.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on an NVIDIA H100 for controllable long-form generation; authored an IEEE-format conference paper.

Emerson
Jun 2022 – Apr 2023

ML Trainee | India

- Developed supervised **regression and classification models** for equipment-failure prediction, anomaly detection, and capacity forecasting on operational data.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification, achieving **89% F1-score** on custom NER tasks.

Suvidha Foundation
Jan 2022 – Mar 2022

ML Intern | India

- Built an end-to-end **multimodal video summarization pipeline** (speech-to-text, transformer-based abstractive summarization, timestamp-aligned clip extraction), deployed across **5,000+ videos** and reducing manual editing effort by **70%**.

PROJECTS

MangaLens
Shipped Chrome extension delivering real-time AI translation of manga and webtoons across 29+ sites — production multimodal AI live on the Chrome Web Store.

Privacy-Preserving Clinical ML Pipeline
Framework combining k-anonymity, l-diversity, and differential privacy to analyze sensitive data without exposing identities, quantifying the privacy–accuracy trade-off.

Story Director LLM
Fine-tuned model generating structured short-form video narratives from a single prompt, paired with an automated rendering pipeline.

EDUCATION

M.S. in Computer Science — *Kent State University, OH* *2025*

GPA: 3.70 / 4.0

Work Authorization: F-1 OPT — authorized to work in the United States now; future sponsorship required.